# Theory Assignment 3
# GNR 638: Machine Learning for Remote Sensing II

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## Problem Statement

Calculate MAP estimation of Gaussian($\mu$, $\sigma$) where prior distribution of $\mu$, $\sigma$ is two different gaussian.

## Proof

## Step-by-Step Solution

### 1. Bayesian Framework

The posterior distribution is proportional to:

$$p(\mu, \sigma|X) \propto p(X|\mu, \sigma)p(\mu)p(\sigma)$$

### 2. Likelihood Function

For independent observations:

$$p(X|\mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

### 3. Prior Distributions

$$p(\mu) = \frac{1}{\sigma_0\sqrt{2\pi}} \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)$$
$$p(\sigma) = \frac{1}{\sigma_\sigma\sqrt{2\pi}} \exp\left(-\frac{(\sigma - \mu_\sigma)^2}{2\sigma_\sigma^2}\right)$$

### 4. Log-Posterior Derivation

Combine components and take logarithm:

$$\begin{aligned}\log p(\mu, \sigma|X) &\propto \log p(X|\mu, \sigma) + \log p(\mu) + \log p(\sigma)\\ &= -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\\ &\quad - \frac{1}{2\sigma_0^2}(\mu - \mu_0)^2 - \frac{1}{2\sigma_\sigma^2}(\sigma - \mu_\sigma)^2 + \text{const}\end{aligned}$$

## 5. MAP Estimation via Partial Derivatives

**For $\mu$:**

Differentiate with respect to $\mu$:

$$\frac{\partial}{\partial \mu} \log p(\mu, \sigma|X) = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu) - \frac{1}{\sigma_0^2}(\mu - \mu_0)$$
$$0 = \frac{n}{\sigma^2}(\bar{x} - \mu) - \frac{1}{\sigma_0^2}(\mu - \mu_0)$$

Solve for $\mu$:

$$\frac{n}{\sigma^2}\mu + \frac{1}{\sigma_0^2}\mu = \frac{n}{\sigma^2}\bar{x} + \frac{1}{\sigma_0^2}\mu_0$$
$$\mu \left( \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) = \frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}$$
$$\mu = \frac{\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$$
$$\Rightarrow \mu = \frac{n\sigma_0^2\bar{x} + \sigma^2\mu_0}{n\sigma_0^2 + \sigma^2}$$

**For $\sigma$:**

Differentiate with respect to $\sigma$:

$$\frac{\partial}{\partial \sigma} \log p(\mu, \sigma|X) = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^{n} (x_i - \mu)^2$$
$$- \frac{1}{\sigma_\sigma^2}(\sigma - \mu_\sigma)$$

Set derivative to zero:

$$-\frac{n}{\sigma} + \frac{S}{\sigma^3} - \frac{\sigma - \mu_\sigma}{\sigma_\sigma^2} = 0$$

where $S = \sum_{i=1}^{n}(x_i - \mu)^2$. Rearranged:

$$\frac{S}{\sigma^3} - \frac{n}{\sigma} = \frac{\sigma - \mu_\sigma}{\sigma_\sigma^2}$$

## 6. Coupled System of Equations

The final MAP estimates satisfy:

$$\mu = \frac{n\sigma_0^2\bar{x} + \sigma^2\mu_0}{n\sigma_0^2 + \sigma^2}$$
$$\frac{\sum(x_i - \mu)^2}{\sigma^3} - \frac{n}{\sigma} = \frac{\sigma - \mu_\sigma}{\sigma_\sigma^2}$$

## 7. Numerical Solution Strategy

1. Initialize with Maximum Likelihood Estimates:

$$\mu^{(0)} = \bar{x}$$
$$\sigma^{(0)} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

2. Iterate until convergence:

   (a) Update $\mu^{(k+1)}$ using current $\sigma^{(k)}$:

   $$\mu^{(k+1)} = \frac{n\sigma_0^2\bar{x} + (\sigma^{(k)})^2\mu_0}{n\sigma_0^2 + (\sigma^{(k)})^2}$$

   (b) Update $\sigma^{(k+1)}$ using current $\mu^{(k+1)}$:

   $$\frac{\sum(x_i - \mu^{(k+1)})^2}{(\sigma^{(k+1)})^3} - \frac{n}{\sigma^{(k+1)}} = \frac{\sigma^{(k+1)} - \mu_\sigma}{\sigma_\sigma^2}$$

   Solve this nonlinear equation using Newton-Raphson method

3. Check convergence:

   $$\|\theta^{(k+1)} - \theta^{(k)}\| < \epsilon$$

## 8. Implementation Considerations

- Handle potential negative $\sigma$ values through constraints
- Use adaptive step sizes in numerical methods
- Monitor log-posterior values for convergence
- Consider parameter transformations for stability